Matthew Woody

matthewwoody.github.io • woodyma@sonoma.edu • linkedin.com/mwoodyy

EDUCATION

Sonoma State University — Expected May 2027
B.S. Applied Mathematics | GPA: 3.41 / 4.00 — Rohnert Park, CA

- *Relevant Coursework:* Partial Differential Equations, Modern Physics, Numerical Analysis, Mathematical Modeling, Linear Algebra, Data Structures, Probability Theory

Santa Rosa Junior College — May 2025
A.S. Mathematics & Natural Sciences — Santa Rosa, CA

RESEARCH EXPERIENCE

Brookhaven National Laboratory — June 2026 – August 2026
Atmospheric Science Intern — Upton, NY
Advisor: Dr. Mindy Deng

- Integrated NEXRAD Level II polarimetric radar data with minute-resolution CoURAGE surface observations to construct an event-based dataset of 18 cold-front passages across heterogeneous terrain and land-cover regimes
- Developed a Python-based tracking and feature-extraction workflow using polarimetric quality masks, connected-component filtering, lag-correlation propagation estimates, and DBSCAN clustering
- Retrieved and composited drop size and rainfall properties to investigate variation with terrain, season, and Chesapeake Bay breeze occurrence, producing inspectable time-range diagrams and spatial profiles
- Identified a recurring mountain-region enhancement in cold front precipitation, characterized by longer duration, broader occurrence of high instantaneous rain rates, and approximately 2-4× greater accumulation than urban region precipitation during the analyzed spring and summer cases
- Project Report (pdf)

University of California, Davis — October 2025 – Present
Undergraduate Atmospheric Science Researcher
Advisor: Dr. Matt Igel

- Built inventory of variations of 7D interacting DoNUT cloud circulations
- Used linear, polynomial and random forest regression to identify broad tendencies between parameterizations

Brookhaven National Laboratory — January 2025 – March 2025
Atmospheric Science Intern — Upton, NY
Advisor: Dr. Mindy Deng

- Applied K-means and DBSCAN clustering to XSACR radar observations from the TRACER campaign to isolate features of Houston's bay breeze
- Developed Python workflows to extract leading-edge position and propagation speed, generate time series statistics and compare frontal velocity with the circulation's overall wind speed

PRESENTATIONS

SSU Annual Math Festival, Rohnert Park, CA, April 2026. "Unsupervised Machine Learning Techniques to Identify Bay Breeze Statistics" (poster).

OTHER EXPERIENCE

SSU Learning and Academic Resource Center August 2026 – Present
Peer Tutor Rohnert Park, CA

- *Courses Supported:* Physics I & II, Differential Equations, Calculus, Reasoning and Proof, Mathematical Modeling, Probability Theory, Descriptive Astronomy, Finite Math (in-class tutor)

Santa Rosa Junior College January 2026 – May 2026
Finite Math Teaching Assistant Santa Rosa, CA

Brookhaven National Laboratory May 2025 – August 2025
Summer Policy Student Upton, NY

City of Petaluma June 2024 – August 2024
Water Resources Intern Petaluma, CA

PROFESSIONAL ORGANIZATIONS

American Meteorological Society (2025 – Present)

TECHNICAL SKILLS

Languages: Python, Octave, C++
Libraries: scikit-learn, Xarray, Matplotlib, NumPy, pandas, NetCDF
Tools: Jupyter Notebook, Excel, LaTeX